

Retrieval-Augmented Generation

Embeddings · Vector Search · Chunking · Advanced RAG

Why RAG?

Hallucination
fluent but wrong

Knowledge cutoff
can't know new facts

No domain depth
generic, not expert

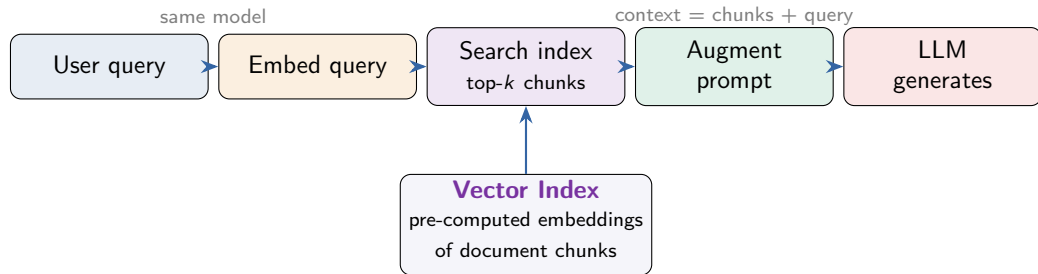
No source citation
"trust me, bro"

↓ **Solution**

RAG: give the LLM access to **external knowledge** at inference time

Lewis et al., "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks," NeurIPS 2020

The Big Picture



Sparse vs Dense Retrieval

BM25 (Sparse)

Bag-of-words: exact term matching

$$\text{score} = \frac{\sum_i \text{IDF}(q_i) \cdot f(q_i, D) \cdot (k_1 + 1)}{f(q_i, D) + k_1(1 - b + b \frac{|D|}{\text{avgdl}})}$$

No training needed, fast

Fails on synonyms/paraphrases

Dense Retrieval (DPR)

Neural embeddings:
semantic matching

$$p_{\eta}(z|x) \propto \exp(q(x)^{\top} d(z))$$

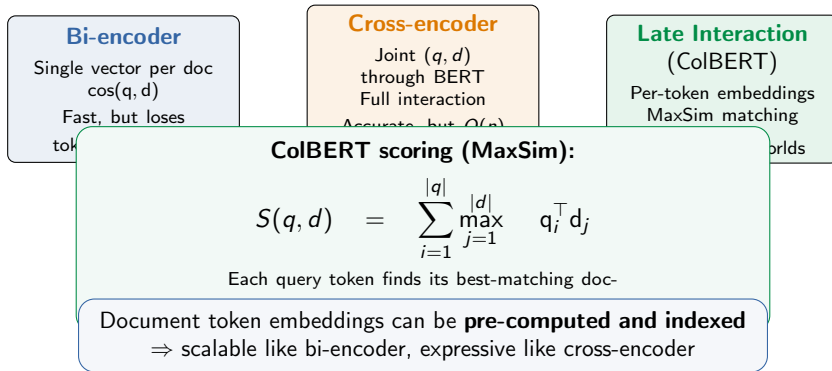
Captures meaning, not just words

Requires training data & compute

Hybrid search = BM25 + Dense →
best of both worlds (Reciprocal Rank Fusion)

Late Interaction: ColBERT

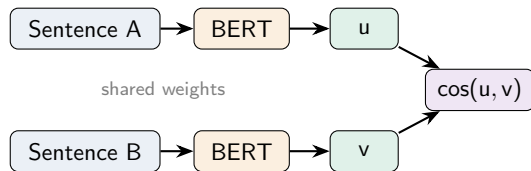
Three retrieval paradigms



Khattab & Zaharia (2020), "ColBERT: Efficient and Effective Passage Search via Late Interaction"

Embedding Models for Retrieval

Sentence-BERT (Reimers & Gurevych, 2019)



Model	Dim	Ctx
ada-002	1536	8K
embed-3-sm	1536	8K
E5-large	1024	512
BGE-M3	1024	8K
Nomic v1	768	8K
Cohere v3	1024	512

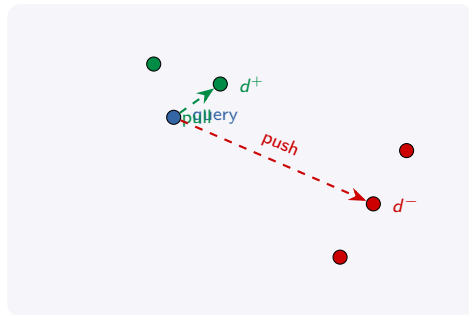
Cosine: $\frac{u \cdot v}{\|u\| \|v\|}$

Dot: $u \cdot v$

L2: $\sqrt{\sum (u_i - v_i)^2}$

Contrastive Training for Embeddings

Embedding Space



InfoNCE loss:

$$\mathcal{L} = -\log \frac{\exp(\text{sim}(q, d^+)/\tau)}{\sum_{j=1}^N \exp(\text{sim}(q, d_j)/\tau)}$$

τ = temperature, N = batch (in-batch negatives)

Hard negatives are critical:
documents that look relevant but aren't

Used in: Sentence-BERT, DPR, SimCSE, CLIP

Vector Databases & ANN Search

1M vectors $\times$ 768 dims = 3 GB — exact
search is $O(n \cdot d)$ per query $\rightarrow$ need **ANN**

HNSW

Hierarchical Navigable
Small World graph
 $O(\log n)$ search
95–99% recall
high memory

IVF

k -means clusters
search nearest centroids
tunable n_{probe}
85–95% recall
moderate memory

Product Quantization

split vector into sub-vectors
quantize each to 1 byte
3072 B $\rightarrow$ 96 B
80–90% recall
very low memory

Vector Database Landscape

FAISS

Meta, library

Pinecone

managed SaaS

Weaviate

hybrid search

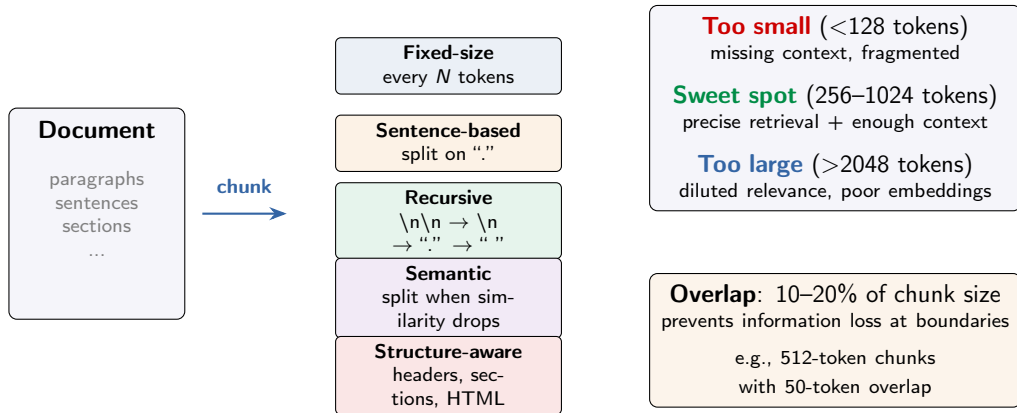
Chroma

lightweight

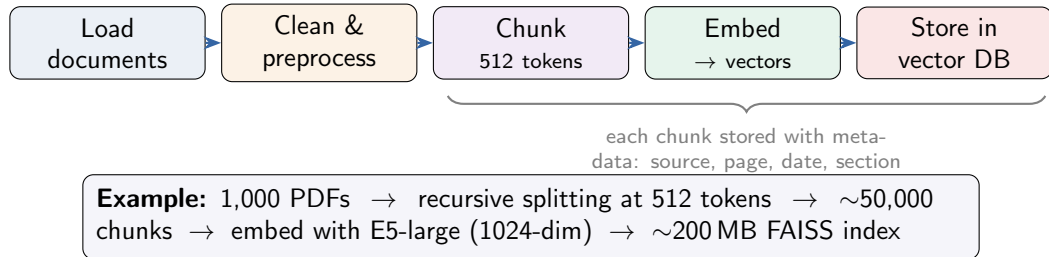
Qdrant

Rust, fast

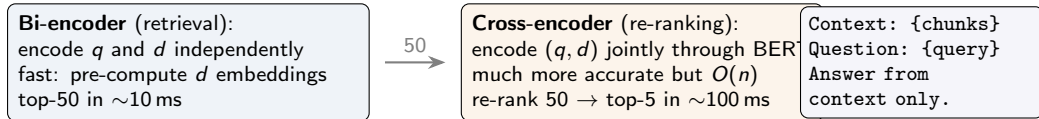
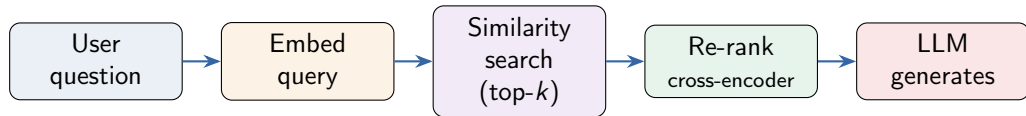
Chunking Strategies



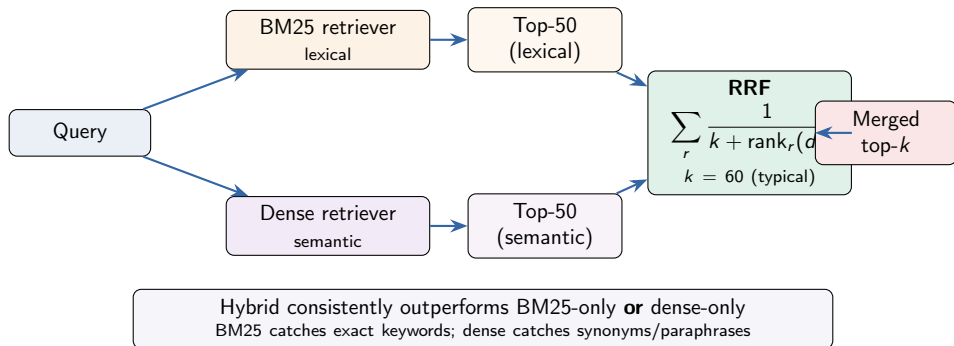
Phase 1: Indexing (Offline)



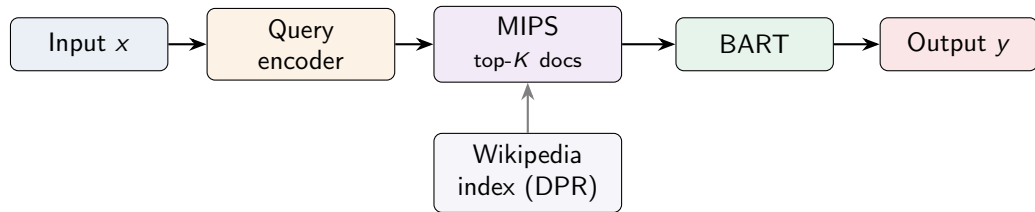
Phase 2: Query (Online)



Hybrid Search & Reciprocal Rank Fusion



The Original RAG Paper (Lewis et al., 2020)



RAG-Sequence:

$$p(y|x) = \sum_z p(z|x) \prod_i p(y_i|x, z, y_{<i})$$

same document for entire sequence

RAG-Token:

$$p(y|x) = \prod_i \sum_z p(z|x) p(y_i|x, z, y_{<i})$$

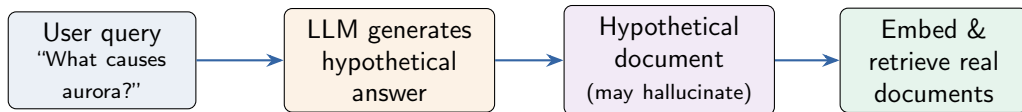
can use different doc per token

Joint end-to-end training of retriever + generator.
SOTA on Natural Questions, TriviaQA, WebQuestions

Query Transformation: HyDE

Problem: query–document mismatch

Short queries embed differently from long document passages



Key insight: the hypothetical doc is in the same “linguistic space” as real documents
paragraph-length, descriptive → better embedding similarity with real passages

Gao et al. (2022). nDCG@10 = 61.3 on TREC DL-20 (vs. 44.5 for Contriever without labels)

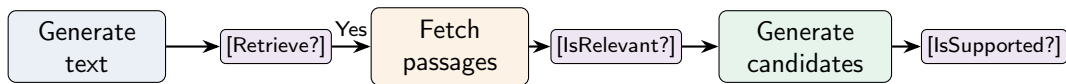
Parent-Child Chunking



Also called “small-to-big retrieval” — search over children, return the parent

Self-RAG (Asai et al., 2023)

Key idea: the model *itself* decides when to retrieve and evaluates what it retrieved



Reflection tokens

[Retrieve]: Yes / No
[IsRel]: relevant?
[IsSup]: supported?
[IsUse]: useful?

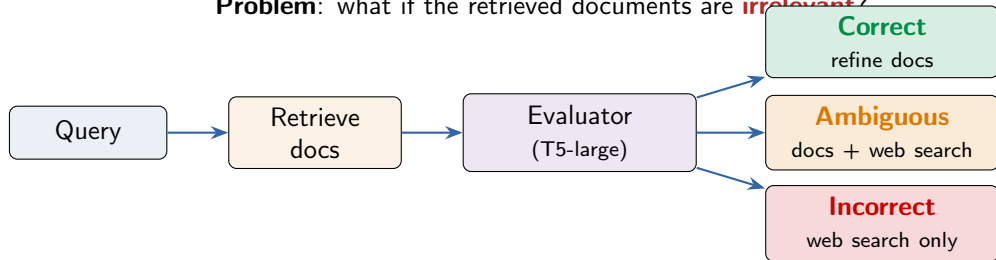
Training:

1. GPT-4 labels data with reflection tokens
2. Fine-tune Llama 2 to predict text + tokens
3. No separate retriever or critic at inference

Result: outperforms vanilla RAG and ChatGPT on factual benchmarks

Corrective RAG (Yan et al., 2024)

Problem: what if the retrieved documents are **irrelevant**?

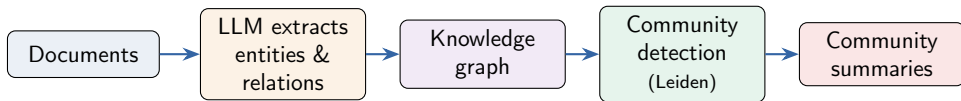


Decompose-then-recompose: split docs into “knowledge strips,” score each for relevance, filter out irrelevant strips, reassemble

CRAG: 61.8% on PopQA (vs. 54.9% for Self-RAG), 86.2 FactScore on biographies

GraphRAG (Edge et al., 2024)

Problem: standard RAG fails on **global queries**
("What are the main themes across all documents?")



Local search:

Entity → neighbors → community context
Good for specific factual questions
Similar to standard RAG but graph-guided

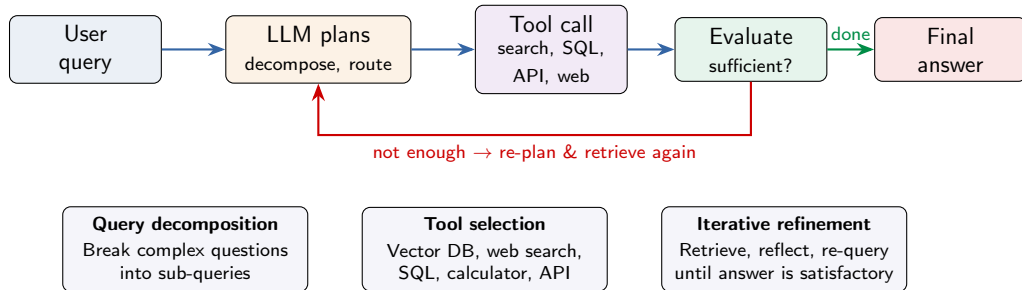
Global search:

Map community summaries → partial answers
Reduce into final synthesized answer
Handles "what are the themes" queries

Structures corpus knowledge as a graph ⇒ enables **multi-hop reasoning** and **global summarization** that flat vector search cannot do

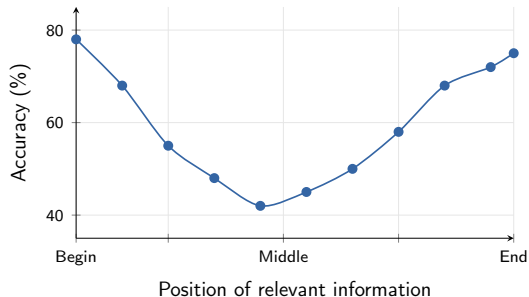
Agentic RAG

Key idea: the LLM acts as an **agent** that decides *when*, *what*, and *how* to retrieve



Frameworks: LangGraph, LlamaIndex Agents, CrewAI · The frontier of RAG research

Lost in the Middle (Liu et al., 2023)



LLMs attend strongly to the **beginning** and **end**

Information in the **middle**
is often **missed**

RAG implication:

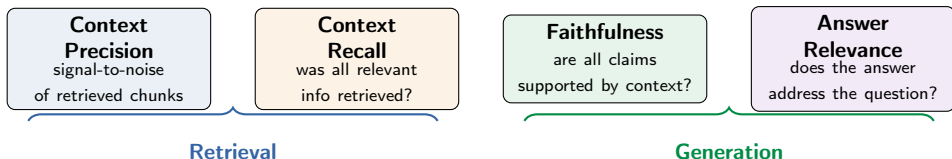
place most relevant chunks
at the **start** or **end** of the prompt

RAG vs Fine-Tuning

	RAG	Fine-Tuning
Knowledge type	facts, up-to-date info	style, format, reasoning
Update knowledge	update the index (easy)	retrain (expensive)
Latency	+100–500 ms overhead	no overhead
Hallucination	reduced (grounded)	can still hallucinate
Source citation	✓ can cite	× no attribution
Setup	vector DB + chunking + embedding	training pipeline + curated data
Best for	factual QA, docs, support	classification, style, extraction

Hybrid (increasingly common): fine-tune for task format + RAG for factual grounding
e.g., fine-tune Llama on medical QA format + RAG over PubMed abstracts

Evaluating RAG: RAGAS Framework



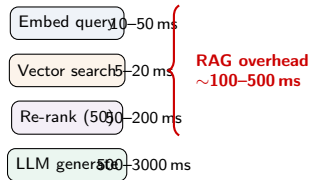
$$\text{Faithfulness} = \frac{\# \text{ claims supported by context}}{\# \text{ total claims in answer}} \quad (\text{LLM-as-judge, no ground truth needed})$$

Common failure modes:

retrieval miss | context overflow | lost in the middle | unfaithful generation

Practical Considerations

Latency Breakdown



Cost

Embedding 1M chunks (OpenAI):
 $512\text{M tokens} \times \$0.02/1\text{M} \approx \$10$

Storage (1M $\times$ 1024-dim):
~4 GB raw, ~8 GB with HNSW

Open-source (E5, BGE):
free (your GPU only)

Index freshness

incremental updates, versioning

Chunk quality

garbage in $\rightarrow$ garbage out

Multi-modal

tables, images, PDFs

Access control

who can see what?

Real-World Applications

Customer
support

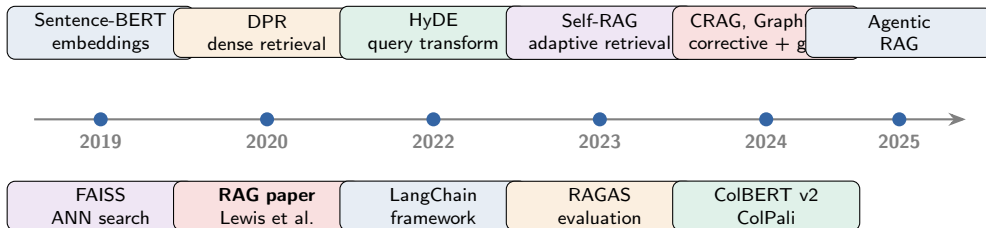
Legal
search

Medical
QA

Code
assistants

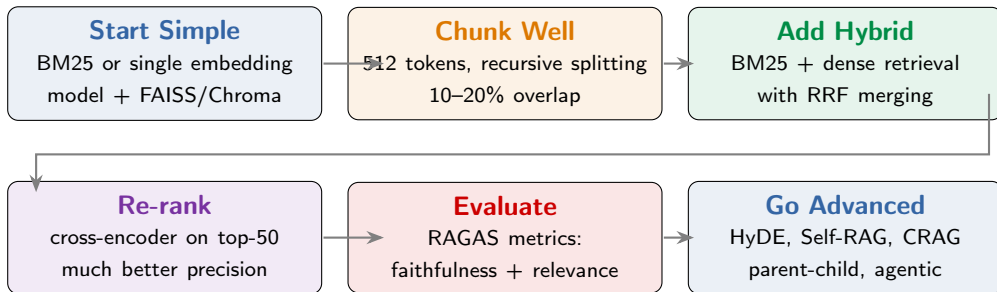
Enterprise
search

RAG Evolution



Trend: simple retrieve-and-read →
adaptive, self-correcting, multi-step, agentic

The RAG Playbook



Don't over-engineer from day one — iterate based on evaluation results

Further reading

Foundations

- Lewis et al. (2020), “Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks”
- Karpukhin et al. (2020), “Dense Passage Retrieval for Open-Domain Question An-

Advanced RAG

- Asai et al. (2024), “Self-RAG: Learning to Retrieve, Generate, and Critique through Self-Reflection”
- Edge et al. (2024), “From Local to Global: A Graph RAG Approach to Query-

Retrieval & Evaluation

- Khattab & Zaharia (2020), “ColBERT: Efficient and Effective Passage Search via Late Interaction”
- Muennighoff et al. (2023), “MTEB: Massive Text Embedding Benchmark”
- Es et al. (2024), “RAGAS: Automated Evaluation of Retrieval Augmented Generation”

Questions?

Next: Hallucination & Grounding